PRACTICAL COURSE WORKBOOK

Cloud Security - AWS & GCP

Apply practical protection to accounts and systems

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a prioritised security review and action plan
SKILLS	AWS, GCP, IAM, Zero Trust, Critical evaluation, Documentation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use AWS appropriately in a realistic, bounded task.
- Use GCP appropriately in a realistic, bounded task.
- Use IAM appropriately in a realistic, bounded task.
- Use Zero Trust appropriately in a realistic, bounded task.

WHO THIS IS FOR

Defenders and developers learning lawful, defensive security practice.

BEFORE YOU BEGIN

- Basic networking and operating-system knowledge

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Cloud Security - AWS & GCP is a structured, practical course covering AWS, GCP, IAM, Zero Trust. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 AWS: Security foundations	1. Assets with AWS 2. Threats with GCP 3. Risk with IAM
02 GCP: Defensive controls	1. Identity with Zero Trust 2. Network protection with AWS 3. Hardening with GCP
03 IAM: Assessment	1. Safe testing with IAM 2. Evidence with Zero Trust 3. Remediation with AWS
04 Zero Trust: Operations	1. Monitoring with GCP 2. Response with IAM 3. Recovery with Zero Trust

HOW TO USE EACH LESSON PAGE

1. Read the objective and identify the decision it supports.
2. Attempt the retrieval prompt before checking the online explanation.
3. Reproduce the worked example with the stated input.
4. Test one normal case and one boundary or failure case.
5. Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

AWS: Security foundations

Apply AWS through assets, threats, risk.

LESSON 01 / 18 MIN

Assets with AWS

Learning objectives

- Explain assets in the context of Cloud Security - AWS & GCP.
- Apply AWS to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms assets, AWS, security

Retrieval prompt In Cloud Security - AWS & GCP, which evidence best supports a assets result produced with AWS?

Evidence to keep [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 02 / 22 MIN

Threats with GCP

Learning objectives

- Explain threats in the context of Cloud Security - AWS & GCP.
- Apply GCP to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms threats, GCP, security

Retrieval prompt In Cloud Security - AWS & GCP, which evidence best supports a threats result produced with GCP?

Evidence to keep [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 03 / 26 MIN

Risk with IAM

Learning objectives

- Explain risk in the context of Cloud Security - AWS & GCP.
- Apply IAM to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms risk, IAM, security

Retrieval prompt In Cloud Security - AWS & GCP, which evidence best supports a risk result produced with IAM?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE
Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY
Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW
Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

GCP: Defensive controls

Apply GCP through identity, network protection, hardening.

LESSON 04 / 30 MIN

Identity with Zero Trust

Learning objectives

- Explain identity in the context of Cloud Security - AWS & GCP.
- Apply Zero Trust to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

identity, Zero Trust, security

Retrieval prompt

In Cloud Security - AWS & GCP, which evidence best supports a identity result produced with Zero Trust?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 05 / 18 MIN

Network protection with AWS

Learning objectives

- Explain network protection in the context of Cloud Security - AWS & GCP.
- Apply AWS to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

network protection, AWS, security

Retrieval prompt

In Cloud Security - AWS & GCP, which evidence best supports a network protection result produced with AWS?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 06 / 22 MIN

Hardening with GCP

Learning objectives

- Explain hardening in the context of Cloud Security - AWS & GCP.
- Apply GCP to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

hardening, GCP, security

Retrieval prompt

In Cloud Security - AWS & GCP, which evidence best supports a hardening result produced with GCP?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

IAM: Assessment

Apply IAM through safe testing, evidence, remediation.

LESSON 07 / 26 MIN

Safe testing with IAM

Learning objectives

- Explain safe testing in the context of Cloud Security - AWS & GCP.
- Apply IAM to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

safe testing, IAM, security

Retrieval prompt

In Cloud Security - AWS & GCP, which evidence best supports a safe testing result produced with IAM?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 08 / 30 MIN

Evidence with Zero Trust

Learning objectives

- Explain evidence in the context of Cloud Security - AWS & GCP.
- Apply Zero Trust to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

evidence, Zero Trust, security

Retrieval prompt

In Cloud Security - AWS & GCP, which evidence best supports a evidence result produced with Zero Trust?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 09 / 18 MIN

Remediation with AWS

Learning objectives

- Explain remediation in the context of Cloud Security - AWS & GCP.
- Apply AWS to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

remediation, AWS, security

Retrieval prompt

In Cloud Security - AWS & GCP, which evidence best supports a remediation result produced with AWS?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Zero Trust: Operations

Apply Zero Trust through monitoring, response, recovery.

LESSON 10 / 22 MIN

Monitoring with GCP

Learning objectives

- Explain monitoring in the context of Cloud Security - AWS & GCP.
- Apply GCP to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms monitoring, GCP, security

Retrieval prompt In Cloud Security - AWS & GCP, which evidence best supports a monitoring result produced with GCP?

Evidence to keep [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 11 / 26 MIN

Response with IAM

Learning objectives

- Explain response in the context of Cloud Security - AWS & GCP.
- Apply IAM to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms response, IAM, security

Retrieval prompt In Cloud Security - AWS & GCP, which evidence best supports a response result produced with IAM?

Evidence to keep [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 12 / 30 MIN

Recovery with Zero Trust

Learning objectives

- Explain recovery in the context of Cloud Security - AWS & GCP.
- Apply Zero Trust to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms recovery, Zero Trust, security

Retrieval prompt In Cloud Security - AWS & GCP, which evidence best supports a recovery result produced with Zero Trust?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE
Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY
Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW
Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Cloud Security - AWS & GCP project using AWS, GCP, IAM.

DELIVERABLE	A working Cloud Security - AWS & GCP artefact plus an evidence-based self-review.
TOOLS	A suitable AWS environment, A plain-text decision log, Test data or realistic sample material

Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using AWS.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

Use security techniques only on systems you own or are explicitly authorised to test.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER
State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS
List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE
Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE
What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION
What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION
Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

Cybersecurity Framework 2.0

NIST - reviewed 2026-08-21

https://www.nist.gov/cyberframework

Web Security Testing Guide

OWASP Foundation - reviewed 2026-08-21

https://owasp.org/www-project-web-security-testing-guide/

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the AWS scope.